STAT 709 Mathematical Statistics Lecture 10 - Fall 2026

Lecture 10: Weak convergence and the central limit theorem

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These notes support STAT 709 at the University of Wisconsin–Madison. Use them alongside the cited textbooks; corrections and questions are welcome.

Mid priority *From consistency to uncertainty*

Overview. Lecture 9 asked whether an estimator approaches its population target. This lecture asks what its remaining error looks like after we put it on a useful scale. We introduce convergence in distribution, state the central limit theorem (CLT), and study three ways to understand its Normal limit: characteristic functions, moments, and Lindeberg's replacement argument. The first gives our complete proof under finite second moments; the other two expose useful ideas for further work.

Quantifying uncertainty of estimation. Consistency says that $\widehat{\theta}_n$ approaches the true parameter θ as $n \to \infty$. But how is $\widehat{\theta}_n$ distributed at a large sample size? Imagine repeating a scientific experiment, or changing the random seed in a computational experiment. We would obtain many realizations of the estimator. Distributional information tells us the size and frequency of its fluctuations, complementing the statement that its error vanishes.

Instructor's Note *Thinking in repeated experiments*

Think of consistency as bringing a camera into focus: the estimates gather around the target. Now zoom in on that gathering cloud. Its shape tells us what an ordinary error looks like and what would count as an unusually lucky or unlucky run. This is a first step toward understanding why two repetitions of an experiment need not give exactly the same answer.

Recap from earlier lectures

From Lectures 3, 5 and 9: convergence and averaging

Reminder only: this block introduces no new exam requirements.

- **Convergence modes**. Almost-sure convergence concerns whole paths; convergence in probability concerns each error tolerance. Used in Section 1.
- **Stochastic orders**. $o_P(1)$ means convergence to zero in probability; $O_P(1)$ means boundedness in probability. Used in Section 1.
- **Weak law**. An i.i.d. sample mean converges to its population mean under a finite first moment. Used in Section 2.
- **Characteristic functions**. The characteristic function of an independent sum is a product. Used in Section 3.1.
- **Dominated convergence**. A justified limit exchange controls the small remainder in the characteristic-function expansion. Used in Section 3.1.

Learning goals

- Read a distributional limit and distinguish it from convergence of random variables on the same probability space.
- Apply the scalar and vector CLT with the correct centering and scale.
- Explain how a quadratic characteristic-function expansion produces an approximately Normal sum.
- Optionally compare moment matching, replacement, and quantitative CDF error bounds.

Priority guide

High Core concepts and calculations to master; directly examined.

Mid Supporting concepts useful in other lectures; in scope for examinations.

Low Enrichment supporting understanding; will not be examined.

Further reading Optional self-study, references, extensions, and historical notes.

A star (*) marks optional material; priority labels give the scope of each block. Lecture-specific exemptions and the optional status of Part VI override the general exam guide.

1 Weak convergence

High priority *Convergence in distribution*

Let X_n and X be random vectors in a fixed dimension d, with joint CDFs F_n and F. For $x \in \mathbb{R}^d$, the inequality $X \le x$ means $X_j \le x_j$ for every coordinate, so $F(x) = \mathbb{P}(X \le x)$.

Definition 1. *We say that F_n **converges weakly** to F, written $F_n \Rightarrow F$ or $F_n \xrightarrow{w} F$, if*

$$F_n(x) \longrightarrow F(x) \quad \text{at every continuity point } x \text{ of } F. \tag{1}$$

*We say that X_n **converges in distribution** to X, written $X_n \xrightarrow{d} X$ or $X_n \Rightarrow X$, when their CDFs have this relationship.*

A continuity point here is a point where the limiting CDF is continuous; F_n itself may have a jump there. In dimension one, requiring convergence only at continuity points lets point masses move toward their limiting location. Requiring convergence at every point would exclude such natural limits.

Exercise prompt. For constant random variables $X_n = 1/n$ and $X = 0$, write their CDFs and check the definition. What happens at the jump of the limiting CDF? See Exercise 1 in Section 6 for the full statement and complete solution.

Remark 1 (Terminology and notation)**.** *"Convergence in distribution," "weak convergence," and "convergence in law" describe the same notion for random vectors. Common notations are*

$$X_n \xrightarrow{d} X, \qquad X_n \xrightarrow{D} X, \qquad X_n \Rightarrow X.$$

Statistics texts often use the first phrase. Weak convergence is also used for probability measures on spaces more general than $\mathbb{R}^d$. The symbol $X \overset{d}{=} Y$ means equality of distributions, not pointwise equality of random variables.

High priority *The distribution matters, not the coupling*

Remark 2. *Unlike convergence in probability or almost surely, convergence in distribution does not require $X, X_1, X_2, \ldots$ to live on the same probability space. It compares their induced probability measures. For $G \sim N(0,1)$, symmetry gives*

$$X_n \Rightarrow G \quad \Longleftrightarrow \quad X_n \Rightarrow -G, \qquad G \overset{d}{=} -G.$$

This statement says nothing about whether $X_n(\omega)$ gets close to $G(\omega)$ for individual outcomes ω.

Instructor's Note *A distribution forgets the labels*

Same pmf under shuffling. Imagine two columns of observations. If we shuffle one column, its histogram stays the same, but its row-by-row relationship with the other column changes. Convergence in distribution compares the histograms; convergence in probability compares paired observations. Intuitively speaking, a distribution remembers how often values occur, but forgets which outcome produced each value.

Exercise prompt. On one probability space let $G \sim N(0,1)$ and $X_n = (-1)^n G$. Does X_n converge in distribution to G? Does it converge in probability to G? See Exercise 2 in Section 6 for the full statement and complete solution.

High priority *Relations with convergence in probability*

Theorem 1. *For random vectors in $\mathbb{R}^d$:*

1. *If $X_n \xrightarrow{p} X$ on a common probability space, then $X_n \Rightarrow X$.*
2. *If $X_n \Rightarrow c$, where $c \in \mathbb{R}^d$ is constant, then $X_n \xrightarrow{p} c$.*
3. *If $X_n \Rightarrow X$, then $X_n = O_P(1)$: for every $\varepsilon > 0$ there are $M < \infty$ and n_0 such that $\mathbb{P}(\|X_n\| > M) < \varepsilon$ whenever $n \geq n_0$.*

In (2), convergence to the constant means concentration of each law near c; the different X_n need not share a probability space.

Proof in dimension one. (1) Compare CDFs with a small horizontal shift. For $\delta > 0$, elementary event inclusions give

$$\begin{aligned} F(x-\delta) - \mathbb{P}(|X_n - X| > \delta) \leq F_n(x) \\ \leq F(x+\delta) + \mathbb{P}(|X_n - X| > \delta). \end{aligned} \tag{2}$$

For example, $X_n \leq x$ and $|X_n - X| \leq \delta$ imply $X \leq x + \delta$. Also, $X \leq x - \delta$ and $|X_n - X| \leq \delta$ imply $X_n \leq x$. First let $n \to \infty$ with δ fixed. Then let $\delta \downarrow 0$ at a continuity point of F. The lower and upper bounds both approach $F(x)$.

(2) A constant limit leaves no probability away from that point. The CDF of c is zero below c and one above c. For every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - c| > \varepsilon) \leq 1 - F_n(c+\varepsilon) + F_n(c-\varepsilon) \longrightarrow 0.$$

The inequality, rather than an equality, accounts for a possible atom of X_n at $c - \varepsilon$.

(3) Keep most of the mass in a fixed bounded interval. Choose $M > 0$ so that F is continuous at both $-M$ and M and $1 - F(M) + F(-M) < \varepsilon/2$. This is possible because the tails vanish and a CDF has only countably many discontinuities; the details are in Exercise 6. For all sufficiently large n,

$$\mathbb{P}(|X_n| > M) \leq 1 - F_n(M) + F_n(-M) < \varepsilon.$$

This is the definition of $O_P(1)$. □

The vector cases. For (1), the same event inclusions use $F(x-\delta\mathbf{1})$ and $F(x+\delta\mathbf{1})$, with error probability $\mathbb{P}(\|X_n - X\|_\infty > \delta)$. At a continuity point of the joint CDF, let $n \to \infty$ and then $\delta \downarrow 0$. For (2), apply the bounded continuous test-function criterion in Theorem 2 to $h(x) = \min(1, \|x-c\|/\varepsilon)$: the expectation tends to $h(c) = 0$ and bounds $\mathbb{P}(\|X_n - c\| > \varepsilon)$. For (3), set $h_M(x) = \min(1, \max(0, \|x\| - M))$. Then $\mathbb{E}\, h_M(X) \le \mathbb{P}(\|X\| > M) \to 0$ as $M \to \infty$, whereas $\mathbb{P}(\|X_n\| > M+1) \le \mathbb{E}\, h_M(X_n) \to \mathbb{E}\, h_M(X)$ for fixed M. Choosing M first and then n proves boundedness in probability.

1.1 Equivalent ways to check weak convergence

High priority *Test functions and characteristic functions*

Let $C_b(\mathbb{R}^d)$ denote the bounded, continuous, real-valued functions on $\mathbb{R}^d$. A **test function** h probes a distribution through $\mathbb{E}\, h(X)$. Boundedness ensures that this expectation exists without an additional moment assumption.

Characteristic functions. The **characteristic function** of a random vector X is

$$\varphi_X(t) = \mathbb{E}\, e^{it^\top X}, \qquad t \in \mathbb{R}^d.$$

It always exists, since $|e^{it^\top X}| = 1$. For a real random variable, $\varphi_X(t) = \mathbb{E}\cos(tX) + i\,\mathbb{E}\sin(tX)$. Independence turns the characteristic function of a sum into the product of the individual characteristic functions, as in Lecture 5.

Theorem 2 (Equivalent criteria for weak convergence)**.** *Let X_n, X be random vectors in $\mathbb{R}^d$. Each of the following is equivalent to $X_n \Rightarrow X$:*

(i) For every $h \in C_b(\mathbb{R}^d)$,

$$\mathbb{E}\, h(X_n) \longrightarrow \mathbb{E}\, h(X). \tag{3}$$

(ii) For every $t \in \mathbb{R}^d$, $\varphi_{X_n}(t) \longrightarrow \varphi_X(t)$.

*(iii) For every $c \in \mathbb{R}^d$, $c^\top X_n \Rightarrow c^\top X$. This is the **Cramér–Wold device**.*

See [2, Theorem 1.9] for the vector statement; [1, Sections 3.2–3.3 and 3.10] gives proofs and further detail. Bounded continuous functions in (i) cannot be replaced by arbitrary unbounded functions: weak convergence alone does not imply convergence of means or variances.

Mid priority *Heuristics for the equivalent criteria*

Instructor's Note *Different ways to observe the same distribution*

Consider a heuristic argument. The CDF is already an expectation: $F(x) = \mathbb{E}\,\mathbf{1}\{X \le x\}$ in dimension one. Its indicator is a sharp on/off measurement. A continuous test function replaces that sharp edge with a soft transition. At a continuity point of F, the thin transition region carries little probability. This is why the sharp and smooth measurements can tell the same limiting story. Exercise 7 works out this picture with two linear ramps.

A characteristic function records oscillating measurements instead. Think of changing t as tuning to a different frequency: together, the frequencies encode the whole distribution. This is a Fourier-transform viewpoint.[1] Taking real and imaginary parts makes the forward implication in (ii) an immediate application of (i); the reverse implication requires a continuity theorem.

Finally, a vector characteristic function already looks in one direction: $\varphi_X(c) = \mathbb{E}\, e^{ic^\top X}$. If every one-dimensional projection has the right limiting law, all these frequency measurements agree. This explains the Cramér–Wold device.

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[1] A Fourier transform describes a function or measure by its responses to oscillations at different frequencies. For a probability measure μ, $\int e^{it^\top x}\, d\mu(x)$ is its characteristic function, with a choice of sign convention. It is the transform of the probability measure, rather than of the CDF viewed as an ordinary integrable function.

2 Central limit theorem

High priority *The Normal limit of a sample mean*

For a random vector $X = (\xi_1, \dots, \xi_d)^\top$, finite second moments ensure that all covariance entries exist. Indeed, $\mathbb{E}\,|\xi_j \xi_k| \le (\mathbb{E}\,\xi_j^2\, \mathbb{E}\,\xi_k^2)^{1/2}$ by Cauchy–Schwarz. Write $\mu = \mathbb{E}\,X_1$ and $\Sigma = \mathrm{Cov}(X_1)$.

Theorem 3 (Central limit theorem)**.** *Let $X_1, X_2, \dots$ be i.i.d. random vectors in a fixed $\mathbb{R}^d$, with $\mathbb{E}\,\|X_1\|^2 < \infty$. Then*

$$\sqrt{n}(\overline{X}_n - \mu) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (X_i - \mu) \Rightarrow N(0, \Sigma).$$

The covariance matrix Σ may be singular. In dimension one, if $\sigma^2 = \mathrm{Var}(X_1) > 0$, the standardized form is

$$\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \Rightarrow N(0, 1).$$

If $\sigma^2 = 0$, then $X_1 = \mu$ almost surely and the centered sum is identically zero almost surely.

Here $N(0, \Sigma)$ can be defined as the law of $\Sigma^{1/2} G$ with $G \sim N(0, I_d)$. Thus a singular covariance causes no difficulty; a full-dimensional density is not required.

What the normalization does. A law of large numbers locates the sample mean near μ. The CLT magnifies its error by $\sqrt{n}$ to retain a nontrivial limiting shape. In dimension one, the unscaled average has variance σ^2/n, whereas $\sqrt{n}(\overline{X}_n - \mu)$ has variance σ^2. For large n, we therefore use $N(\mu, \sigma^2/n)$ as an approximation to the sampling law of $\overline{X}_n$. The theorem concerns these rescaled errors; it does not say that the individual observations become Normal.

Exercise prompt. For i.i.d. Bernoulli observations with a fixed success probability $p \in (0, 1)$, identify the centering and scaling for a standard Normal limit of the sample proportion. What changes when $p = 0$ or $p = 1$? See Exercise 3 in Section 6 for the full statement and complete solution.

Mid priority *Why the CLT appears so often*

Instructor's Note *Many small contributions*

Imagine a measurement error built from many small, independent pushes. Each push may have an irregular shape, but the sum gradually loses those details. After centering and rescaling, the mean and covariance are what remain visible. This is the striking part of the CLT: many different microscopic mechanisms lead to the same macroscopic shape.

The same averaging idea helps explain why Gaussian objects arise in Brownian motion,[2] diffusion and the heat equation, and some models of financial fluctuations. Those are further directions; here our focus is the sampling distribution of a sum or an average.

Mid priority *How sums approach a Normal distribution*

Let X_i independently equal -1 or 1, each with probability $1/2$. For $S_n = n^{-1/2} \sum_i X_i$, counting the positive signs gives

$$\mathbb{P}\left(S_n = \frac{2k - n}{\sqrt{n}}\right) = \binom{n}{k} 2^{-n}, \qquad k = 0, \dots, n.$$

[2] Standard Brownian motion is a continuous random path $B(t)$, $t \ge 0$, starting at zero, with independent increments and $B(t) - B(s) \sim N(0, t - s)$ for $t > s$. It models accumulated random fluctuations over time. Process-level convergence to such paths requires more than the scalar CLT stated here.

The mean is zero and the variance is one for every n. For $n = 4$, the complete distribution is

s	-2	-1	0	1	2
$\mathbb{P}(S_4 = s)$	$1/16$	$4/16$	$6/16$	$4/16$	$1/16$

Reading the comparison. As n grows, the possible values become more closely spaced, and the step CDF approaches the standard Normal CDF at every real argument. Each finite distribution is still discrete. If we draw bars on intervals of width $2/\sqrt{n}$, their heights must be probability divided by this width so that their areas, rather than their heights, represent mass. A smooth bell curve over those bars is a comparison with a density, not an assertion that the discrete sum has a density. The CDF comparison puts both laws on the same probability scale.

3 Three proof ideas for the CLT

High priority *Reduction to centered scalar variables*

The Cramér–Wold device reduces the vector statement to scalar projections. For each fixed $c \in \mathbb{R}^d$, the required limit is

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} c^\top (X_i - \mu) \Rightarrow N(0, c^\top \Sigma c).$$

When the projected variance is positive, divide by its square root and prove the unit-variance scalar theorem. A zero projected variance gives a constant projection almost surely. Translation and multiplication by fixed constants preserve weak convergence; Exercise 9 gives a short proof with bounded continuous test functions.

Exercise prompt. Justify that $c^\top X_1$ has a finite second moment and variance $c^\top \Sigma c$. Explain the zero-variance case without dividing by zero. See Exercise 4 in Section 6 for the full statement and complete solution.

3.1 Characteristic functions

Mid priority *From products to a quadratic limit*

Recall the characteristic function and its product rule. A probability law is uniquely determined by this function; see [1, Theorem 3.3.11]. The following result also allows us to identify a law from a limiting characteristic function.

Theorem 4 (Lévy's continuity theorem)**.** *Let μ_n be probability measures on $\mathbb{R}$, with characteristic functions φ_n.*

1. *If $\mu_n \Rightarrow \mu$, then $\varphi_n(t) \to \varphi_\mu(t)$ for every $t \in \mathbb{R}$.*
2. *If $\varphi_n(t) \to \varphi(t)$ for every $t \in \mathbb{R}$ and φ is continuous at zero, then φ is the characteristic function of a probability measure μ, and $\mu_n \Rightarrow \mu$.*

The proof of this continuity theorem is optional reading in [1, Theorem 3.3.17]. Its statement is the tool used below; continuity at zero must not be dropped when a limit law is not already known.

Proof of the scalar CLT. Suppose $\mathbb{E}\, X_1 = 0$ and $\mathbb{E}\, X_1^2 = 1$. Write $S_n = n^{-1/2} \sum_{i=1}^{n} X_i$ and $\psi(t) = \mathbb{E}\, e^{itX_1}$.

Step 1: justify the quadratic expansion. For real u, set $r(u) = e^{iu} - 1 - iu + u^2/2$. Taylor's formula gives $r(u)/u^2 \to 0$ as $u \to 0$, with the ratio defined as zero at zero. The integral remainder for the linear Taylor

polynomial gives $|e^{iu} - 1 - iu| \le u^2/2$, so $|r(u)| \le u^2$ for all real u. Consequently, $|r(tX_1)|/t^2 \le X_1^2$ and DCT yields

$$\psi(t) = 1 + it\,\mathbb{E}\,X_1 - \frac{t^2}{2}\,\mathbb{E}\,X_1^2 + \mathbb{E}\,r(tX_1) = 1 - \frac{t^2}{2} + o(t^2). \tag{4}$$

Only a finite second moment was used. No third-moment bound is hidden in this step. Exercise 8 also checks directly that $\psi'(0) = i\,\mathbb{E}\,X_1$ and $\psi''(0) = -\,\mathbb{E}\,X_1^2$.

Step 2: combine independent summands. For each fixed t, independence gives

$$\varphi_{S_n}(t) = \psi(t/\sqrt{n})^n = \left(1 - \frac{t^2}{2n} + o(n^{-1})\right)^n.$$

Here the remainder is evaluated with t fixed and $n \to \infty$. To justify the limit even though ψ can be complex, let $a_n = \psi(t/\sqrt{n})$ and $b_n = e^{-t^2/(2n)}$. Both have modulus at most one, and $a_n - b_n = o(n^{-1})$. Factorization gives

$$|a_n^n - b_n^n| \le |a_n - b_n| \sum_{j=0}^{n-1} |a_n|^j |b_n|^{n-1-j} \le n|a_n - b_n| \longrightarrow 0.$$

Since $b_n^n = e^{-t^2/2}$, we obtain $\varphi_{S_n}(t) \to e^{-t^2/2}$. This is the characteristic function of $N(0,1)$ and is continuous at zero. Lévy's theorem completes the proof. The reduction above gives the general scalar and vector cases. □

Instructor's Note *Why the quadratic term survives*

Consider the expansion as a bookkeeping device. Centering removes the linear term. Each remaining quadratic contribution is of size $1/n$, and there are n contributions, so it survives at order one. The smaller terms disappear. The exponent of a Gaussian characteristic function is exactly quadratic, which is why the Normal law fits the calculation so naturally.

For a centered, unit-variance summand Z, the characteristic-function expansion is

$$\mathbb{E}\,e^{itZ/\sqrt{n}} = 1 + \underbrace{\frac{it}{\sqrt{n}}\,\mathbb{E}\,Z}_{0} - \underbrace{\frac{t^2}{2n}\,\mathbb{E}\,Z^2}_{t^2/(2n)} + \underbrace{o(n^{-1})}_{\text{remainder}}.$$

3.2 The moment method (optional)

Low priority *When moments determine a limiting law*

The **moment method** studies $\mathbb{E}\,X_n^k$ for successive integers k. This is a proof strategy, not the method of moments for constructing estimators in Lecture 8. It can also be useful for dependent models and random matrices, although the counting is often more involved. Moments do not determine every probability law. We therefore need a condition that controls tails and makes moment information sufficient.

Definition 2. *A collection $(Z_i)_{i\in\mathcal{I}}$ is **uniformly subgaussian** if there are constants $C, c > 0$, independent of i, such that*

$$\mathbb{P}(|Z_i| \ge t) \le Ce^{-ct^2} \quad \text{for every } t > 0 \text{ and every } i \in \mathcal{I}.$$

A single variable satisfying such a bound is called subgaussian.

This definition controls the uncentered absolute value. Uniformity means one pair of constants works for the whole collection. Gaussian tail decay is a useful model for such light tails.

Theorem 5 (Moment convergence under uniform subgaussian tails)**.** *Suppose (Z_n) is uniformly subgaussian and Z is subgaussian. Then*

$$Z_n \Rightarrow Z \quad \Longleftrightarrow \quad \mathbb{E}\, Z_n^k \longrightarrow \mathbb{E}\, Z^k \text{ for every integer } k \geq 0.$$

This is a convenient sufficient version of the moment method. More general results use moment determinacy, meaning uniqueness of the law specified by its moments, and conditions such as Carleman's; see [1, Section 3.3.5].

Proof. Weak convergence implies moment convergence. Choose a continuous $\chi : \mathbb{R} \to [0,1]$ equal to one on $[-1,1]$ and zero outside $[-2,2]$. For a fixed $N > 0$, $x^k \chi(x/N)$ is bounded and continuous, so

$$\mathbb{E}[Z_n^k \chi(Z_n/N)] \longrightarrow \mathbb{E}[Z^k \chi(Z/N)]. \tag{5}$$

For a nonnegative Y and $k \geq 1$, Tonelli's theorem gives the exact truncated-tail identity

$$\mathbb{E}[Y^k \mathbf{1}\{Y > N\}] = N^k\, \mathbb{P}(Y > N) + \int_N^\infty k t^{k-1}\, \mathbb{P}(Y > t)\, dt. \tag{6}$$

In particular, the boundary term $N^k\, \mathbb{P}(Y > N)$ is part of this expression. The subgaussian bound now yields

$$\begin{aligned} \sup_n \mathbb{E}[|Z_n|^k \mathbf{1}\{|Z_n| > N\}] &\leq C N^k e^{-cN^2} + C \int_N^\infty k t^{k-1} e^{-ct^2}\, dt \\ &\longrightarrow 0 \qquad (N \to \infty). \end{aligned} \tag{7}$$

It also implies that all moments of Z are finite, and DCT gives

$$\mathbb{E}[|Z|^k (1 - \chi(Z/N))] \longrightarrow 0. \tag{8}$$

Bound the difference of the untruncated moments by the two tails and the difference in (5). First let $n \to \infty$ for fixed N, then let $N \to \infty$. The case $k = 0$ is immediate.

Moment convergence implies weak convergence. Enlarge C and decrease c if necessary so that the same tail bound holds for every Z_n and for Z. Integrating the tail at even orders gives

$$\mathbb{E}\,|Z_n|^{2m},\ \mathbb{E}\,|Z|^{2m} \leq C \int_0^\infty 2m t^{2m-1} e^{-ct^2}\, dt = C c^{-m} m!.$$

For $P_{2m-1}(u) = \sum_{j=0}^{2m-1} (iu)^j / j!$, the Taylor remainder along the real line satisfies $|e^{iu} - P_{2m-1}(u)| \leq |u|^{2m}/(2m)!$. Thus, for fixed t,

$$\limsup_{n \to \infty} |\varphi_{Z_n}(t) - \varphi_Z(t)| \leq \frac{2C(t^2/c)^m m!}{(2m)!}.$$

Indeed, the expectations of the finite polynomial converge by the moment assumption, and the two remainders have the stated bound. The right side tends to zero: the ratio of successive terms is $(t^2/c)(m+1)/[(2m+2)(2m+1)] \to 0$. Lévy's theorem proves the claim. □

Low priority *Moments of a normalized sum*

For this proof assume that the i.i.d. variables satisfy $|X_i| \leq M$, $\mathbb{E}\, X_i = 0$, and $\mathbb{E}\, X_i^2 = 1$. This is stronger than the CLT's assumptions. The normalized sums $S_n = n^{-1/2} \sum_i X_i$ are uniformly subgaussian; Exercise 11 proves this directly. The previous theorem therefore reduces the bounded-variable CLT to checking $\mathbb{E}\, S_n^k \to \mathbb{E}\, G^k$ for every k, where $G \sim N(0,1)$. The finite-second-moment case has already been proved by characteristic functions, so it does not depend on an unstated truncation argument here.

Warm-up calculations. The first three orders are $\mathbb{E}\, S_n^0 = 1$, $\mathbb{E}\, S_n = 0$, and $\mathbb{E}\, S_n^2 = 1$. At the third order, expand according to repeated indices:

$$\mathbb{E}\, S_n^3 = n^{-3/2}\Bigg\{ \sum_i \mathbb{E}\, X_i^3 + 3\sum_{i<j} \mathbb{E}(X_i^2 X_j + X_i X_j^2) + 6 \sum_{i<j<\ell} \mathbb{E}(X_i X_j X_\ell)\Bigg\} = n^{-1/2}\, \mathbb{E}\, X_1^3 \longrightarrow 0.$$

Independence and zero means eliminate every term in which some index appears exactly once.

Exercise prompt. Under these bounded, centered, unit-variance assumptions, calculate $\mathbb{E}\, S_n^4$ exactly in terms of n and $\mathbb{E}\, X_1^4$. Identify the index patterns that survive. See Exercise 5 in Section 6 for the full statement and complete solution.

General expansion. For a fixed positive integer k, group the terms by the number m of distinct indices and their positive multiplicities:

$$\mathbb{E}\, S_n^k = n^{-k/2} \sum_{m=1}^{k} \sum_{1 \le i_1 < \cdots < i_m \le n} \sum_{\substack{a_1 + \cdots + a_m = k \\ a_j \ge 1}} \frac{k!}{a_1! \cdots a_m!} \prod_{j=1}^{m} \mathbb{E}\, X_{i_j}^{a_j}. \tag{9}$$

If any $a_j = 1$, the term is zero. For a surviving pattern, $a_j \ge 2$, so $m \le \lfloor k/2 \rfloor$. There are $\binom{n}{m} = O(n^m)$ choices of indices, and their moments are bounded for this fixed k. Every group with $m < k/2$ is therefore $o(1)$ after normalization. If $k = 2r$ is even, the only order-one pattern has $m = r$ and every $a_j = 2$, giving

$$\mathbb{E}\, S_n^{2r} = n^{-r} \binom{n}{r} \frac{(2r)!}{2^r} + o(1) \longrightarrow \frac{(2r)!}{2^r r!}.$$

If k is odd, every group is $o(1)$ and $\mathbb{E}\, S_n^k \to 0$. These are precisely the Gaussian moments: symmetry gives the odd moments, and integration by parts against the Normal density gives $\mathbb{E}\, G^{2r} = (2r-1)\, \mathbb{E}\, G^{2r-2}$ starting from $\mathbb{E}\, G^0 = 1$. This proves the CLT under the boundedness assumption.

Instructor's Note *Pairings explain the square-root scale*

There are n^k terms before we take the expectation. Most disappear: a lone occurrence of an index brings in a zero mean. The largest surviving families pair the indices, leaving about $n^{k/2}$ contributions. Taking a kth power of the normalization contributes exactly $n^{-k/2}$. This cancellation is the combinatorial reason for the square-root scale. You can picture the terms as groups of positions carrying the same index; pairs are the groups that survive at leading order.

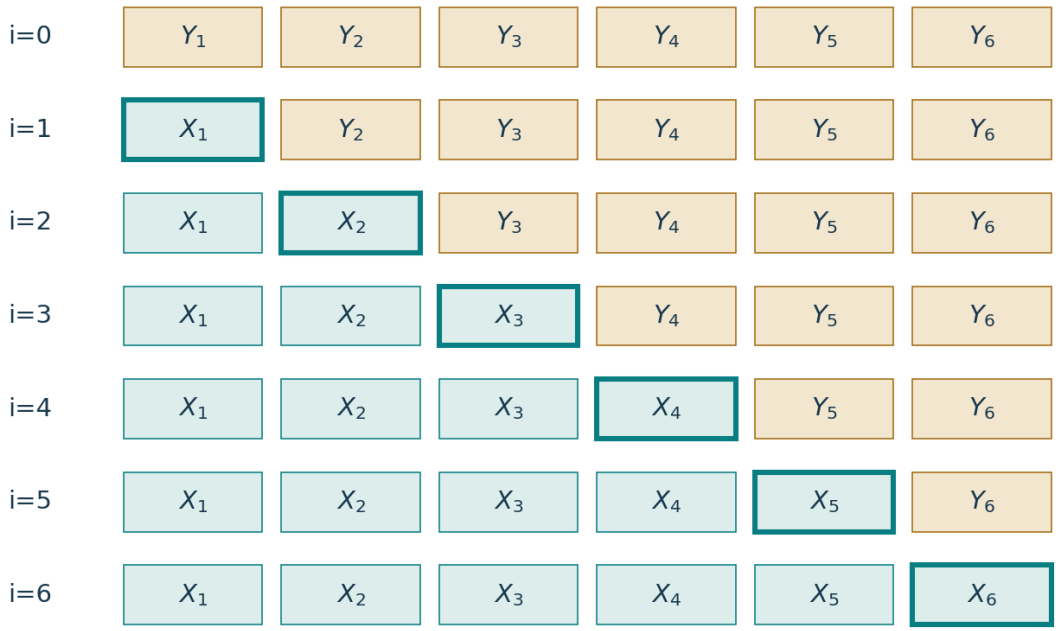
3.3 Lindeberg's replacement argument

Low priority *Replace one observation at a time*

The third proof idea is closely related to **universality**: the same limiting behavior can arise from different input distributions. One first compares a statistic for general inputs with the same statistic for Gaussian inputs, and then calculates in the Gaussian case. The comparison is analytic; the Gaussian calculation is often simpler algebra.

Instructor's Note *A bridge made of small replacements*

Replacing every observation at once seems like a big change. Instead, build a bridge of n small steps: replace the first observation by a Gaussian, then the second, and so on. If one replacement changes a smooth measurement by only order $n^{-3/2}$, the whole bridge costs order $n^{-1/2}$. Matching the mean and variance removes the two larger terms. This is a useful research habit: separate the comparison argument from the calculation in the model where the answer is easiest.



Each row changes exactly one term; the outlined cell is the term just replaced. All seven rows are shown.

We make the comparison precise using a smooth test function. Let $\mathcal{F}_B$ be the functions $\phi \in C^3(\mathbb{R})$ with $\|\phi^{(j)}\|_\infty \le B$ for $j = 0, 1, 2, 3$, where $\phi^{(0)} = \phi$. Such functions belong to $C_b(\mathbb{R})$; we additionally control the derivatives needed in Taylor's formula.

Theorem 6 (A bound for smooth test functions)**.** *Let $X_1, X_2, \dots$ be i.i.d., with $\mathbb{E}\, X_1 = 0$, $\mathbb{E}\, X_1^2 = \sigma^2$, and $\rho = \mathbb{E}\,|X_1|^3 < \infty$. Define $S_n = n^{-1/2} \sum_i X_i$ and $G_\sigma \sim N(0, \sigma^2)$. Then*

$$\sup_{\phi \in \mathcal{F}_B} |\,\mathbb{E}\,\phi(S_n) - \mathbb{E}\,\phi(G_\sigma)| \le \frac{B}{6\sqrt{n}}\big(\rho + \mathbb{E}\,|G_\sigma|^3\big) \le \frac{CB\rho}{\sqrt{n}},$$

where $C = (1 + 2\sqrt{2/\pi})/6$ is a numerical constant.

Proof. Take independent $Y_i \sim N(0, \sigma^2)$, jointly independent of all X_i. Put

$$S_{n,i} = \frac{X_1 + \cdots + X_i + Y_{i+1} + \cdots + Y_n}{\sqrt{n}}, \quad i = 0, \dots, n.$$

The endpoints are $S_{n,n} = S_n$ and $S_{n,0} \overset{d}{=} G_\sigma$. For $i = 0, \dots, n-1$, let $T_{n,i} = (X_1 + \cdots + X_i + Y_{i+2} + \cdots + Y_n)/\sqrt{n}$. Then $S_{n,i} = T_{n,i} + Y_{i+1}/\sqrt{n}$ and $S_{n,i+1} = T_{n,i} + X_{i+1}/\sqrt{n}$.

Match the first two Taylor terms. For $V = X_{i+1}$ or $V = Y_{i+1}$, the third-order remainder bound gives

$$\begin{aligned}\mathbb{E}\,\phi(T_{n,i} + V/\sqrt{n}) = \mathbb{E}\,\phi(T_{n,i}) + \frac{\mathbb{E}\,\phi'(T_{n,i})\,\mathbb{E}\,V}{\sqrt{n}} \\ + \frac{\mathbb{E}\,\phi''(T_{n,i})\,\mathbb{E}\,V^2}{2n} + R_V, \qquad |R_V| \le \frac{B\,\mathbb{E}\,|V|^3}{6n^{3/2}}.\end{aligned} \tag{10}$$

Independence of $T_{n,i}$ and V justifies the products of expectations. The linear terms vanish and the quadratic terms agree, since both choices of V have mean zero and variance σ^2. Hence

$$|\,\mathbb{E}\,\phi(S_{n,i+1}) - \mathbb{E}\,\phi(S_{n,i})| \le \frac{B}{6n^{3/2}}\big(\rho + \mathbb{E}\,|G_\sigma|^3\big).$$

Add the replacement errors. Sum these n bounds in the telescoping identity

$$\mathbb{E}\,\phi(S_n) - \mathbb{E}\,\phi(S_{n,0}) = \sum_{i=0}^{n-1}\big[\,\mathbb{E}\,\phi(S_{n,i+1}) - \mathbb{E}\,\phi(S_{n,i})\big].$$

Finally, $\mathbb{E}\,|G_\sigma|^3 = 2\sqrt{2/\pi}\,\sigma^3$ and $\sigma^3 \le \rho$ by the moment inequality, giving the second bound. If $\sigma = 0$, both inputs are zero almost surely and the conclusion holds with zero error. □

For a fixed t, both $\cos(tx)$ and $\sin(tx)$ lie in some $\mathcal{F}_B$, for example with $B = \max(1, |t|^3)$. Their expectation differences therefore vanish. The characteristic-function criterion gives a CLT under this argument's stronger third-moment assumption.

Low priority *The Berry–Esseen bound*

A sharper theorem measures the discrepancy between the entire CDFs, rather than between smooth test functions with a fixed derivative bound. Let Φ denote the standard Normal CDF.

Theorem 7 (Berry–Esseen theorem)**.** *Under the preceding i.i.d. assumptions, suppose $\sigma > 0$ and $\rho = \mathbb{E}\,|X_1|^3 < \infty$. There is a universal finite constant C_{BE} such that*

$$\sup_{a\in\mathbb{R}}\left|\mathbb{P}\left(\frac{\sum_{i=1}^n X_i}{\sigma\sqrt{n}} \le a\right) - \Phi(a)\right| \le \frac{C_{\mathrm{BE}}\rho}{\sigma^3\sqrt{n}}.$$

Equivalently, compare $S_n = \sum_i X_i/\sqrt{n}$ with $N(0,\sigma^2)$, using $\Phi(a/\sigma)$ in place of $\Phi(a)$.

The standardized third absolute moment ρ/σ^3 is unchanged when the observations are multiplied by a nonzero constant. A bound without this normalization cannot hold uniformly over scales. The theorem implies the CLT under its additional third-moment assumption and supplies a uniform finite-sample CDF error bound. It need not be small for a particular n, and it gives no density approximation. See [1, Section 3.4.4] for a proof.

The smooth bound above is not, by itself, a proof of the same $n^{-1/2}$ CDF rate: approximating an indicator by a narrow smooth transition makes its derivatives large. The Berry–Esseen proof needs additional control of this approximation. Exercise 12 checks the scale factors explicitly.

Key ideas

- Weak convergence concerns distributions, not a chosen pairing of observations.
- The CLT needs the correct centering, normalization and assumptions.
- Bounded-continuous tests, distribution functions and characteristic functions offer different checks for weak convergence.
- Convergence in distribution alone does not justify convergence of moments.

4 Detailed reference

High priority *A guide to the main definitions and results*

The table links back to each definition or statement. High and mid material are in exam scope; low material is optional enrichment. The priority in each row belongs to that item.

Notion or result	What to remember	Tier
Weak convergence	CDF convergence at continuity points of the limiting CDF.	High
Law versus paired variables	Only distributions matter; a common probability space is unnecessary.	High
Probability and distribution limits	Probability convergence implies weak convergence; a constant weak limit implies concentration.	High
Boundedness in probability	A weakly convergent sequence is $O_P(1)$.	High
Bounded continuous tests	Compare expectations of every bounded continuous function.	High
Characteristic function	$\varphi_X(t) = \mathbb{E}\, e^{it^\top X}$; independent sums give products.	High
Cramér–Wold	Every scalar projection determines the vector limit.	High
Central limit theorem	Center i.i.d. sums and divide by $\sqrt{n}$; assume finite second moments.	High
Singular covariance	Zero variance in a direction gives a constant projection.	High
Lévy continuity theorem	A pointwise characteristic-function limit continuous at zero is the characteristic function of a probability law.	Mid
Quadratic expansion	Centering cancels the linear term; a DCT remainder needs only a second moment.	Mid
Uniform subgaussian tails	One Gaussian-type tail bound works for the whole collection.	Low
Moment convergence	Uniform tail control connects unbounded moments to weak convergence.	Low
Pairings	Singleton indices vanish; paired indices give Gaussian leading moments.	Low
Lindeberg replacement	Match two moments; sum the third-order replacement errors.	Low
Berry–Esseen	Uniform CDF error is at most a constant times $\rho/(\sigma^3\sqrt{n})$.	Low
Scheffé's theorem	Almost-everywhere convergence of normalized densities gives L^1 convergence.	Low

Looking ahead

Lecture 11 transfers CLT approximations through transformations of an estimator. Continuous mapping, Slutsky and the delta method turn the convergence tools here into practical asymptotic calculations. Continue to Lecture 11.

5 Further reading

Further reading *References and useful measure facts*

For weak convergence, characteristic functions, moment determinacy and CLTs, consult [1, Sections 3.2–3.4 and 3.10]; the author's fifth-edition book PDF is available online. Shao's [2, Theorem 1.9] is a standard reference for the equivalent vector criteria. The three proof approaches follow the themes of [3, Section 2.2]; Tao's course notes on the CLT provide a companion discussion.

Some measure facts that support these arguments are worth recording:

1. A one-dimensional CDF has at most countably many discontinuities. These are precisely the atoms x with $\mathbb{P}(X = x) > 0$. Exercise 6 gives the counting argument.

2. If B is a Borel subset of $\mathbb{R}$ and $m(B) < \infty$, then for any $\varepsilon > 0$ there are a compact $K \subseteq B$ and an open $O \supseteq B$ with $m(B \setminus K) < \varepsilon$ and $m(O \setminus B) < \varepsilon$. This is regularity of Lebesgue measure. For infinite-measure sets, formulate approximation locally on bounded intervals rather than subtracting ε from infinity. See Tao's *An Introduction to Measure Theory*, Section 1.2 for the inner and outer regularity of Lebesgue measure.

3. Every open subset of $\mathbb{R}$ is a countable union of disjoint open intervals, allowing unbounded endpoints. Its interval components each contain a distinct rational number, which establishes countability.

Low priority *A useful criterion for densities*

The following theorem also covers probability mass functions by using counting measure. Pointwise convergence of CDFs generally says nothing about pointwise convergence of their densities, so this is a separate sufficient condition.

Theorem 8 (Scheffé's theorem)**.** *Let f_n and f be nonnegative densities with respect to a common measure ν, with $\int f_n \, d\nu = \int f \, d\nu = 1$. If $f_n \to f$ ν-almost everywhere, then*

$$\int |f_n - f| \, d\nu \longrightarrow 0.$$

In particular, probabilities of measurable sets converge uniformly: $|\int_A (f_n - f) \, d\nu| \le \int |f_n - f| \, d\nu$. This is stronger than weak convergence and is related to total variation distance.[3] Exercise 13 gives a complete proof and explains why the limiting density must still integrate to one.

6 Exercises and complete solutions

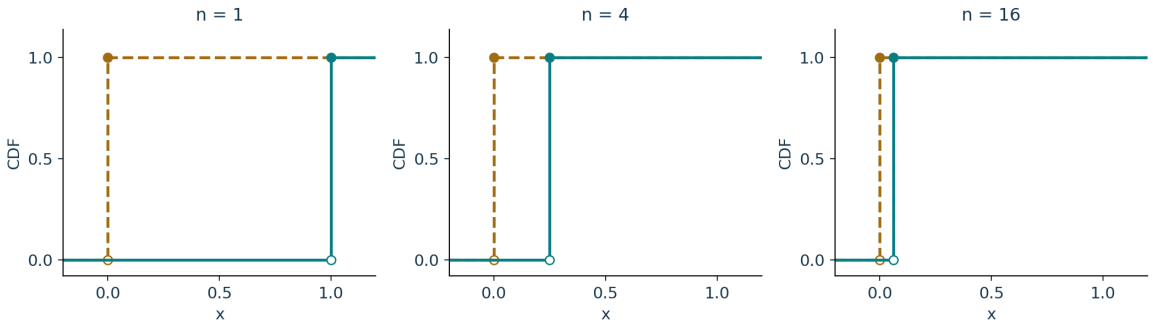
Try each problem before reading its solution. The five short prompts in the main text point to Exercises 1–5 below. Further exercises supply proof details and useful extensions; each retains its own priority.

High priority *Exercise 1: Moving point masses*

Exercise 1. Let $X_n = 1/n$ and $X = 0$ almost surely. Write F_n, F and verify $X_n \Rightarrow X$. Does $F_n(0)$ converge to $F(0)$? Extend the argument to constants $a_n \to a$.

Solution. We have $F_n(x) = \mathbf{1}\{x \ge 1/n\}$ and $F(x) = \mathbf{1}\{x \ge 0\}$. For fixed $x < 0$, $F_n(x) = F(x) = 0$. For fixed $x > 0$, eventually $1/n \le x$, so $F_n(x) = F(x) = 1$. These are all continuity points of F, proving weak convergence. At zero, $F_n(0) = 0$ for every n but $F(0) = 1$; the definition does not require convergence at this jump. For $a_n \to a$, any fixed $x < a$ eventually lies below a_n, and any fixed $x > a$ eventually lies above a_n. The same argument proves $\delta_{a_n} \Rightarrow \delta_a$. □

[3] For probability laws with densities f, g relative to a common measure, the total variation distance is $\sup_A |\int_A (f - g) \, d\nu| = \frac{1}{2} \int |f - g| \, d\nu$. It measures the largest difference in probability assigned to the same measurable event.



The jumps at $1/n$ move toward zero. At zero the CDFs never agree with the limit; at every continuity point of the limit they converge.

High priority *Exercise 2: Equal laws need not give close variables*

Exercise 2. Let $G \sim N(0,1)$ and $X_n = (-1)^n G$ on the same probability space. Check distributional convergence to G and decide whether convergence in probability to G holds.

Solution. Symmetry gives $X_n \overset{d}{=} G$ for every n, so the CDFs are identical and $X_n \Rightarrow G$. For odd n, however, $|X_n - G| = 2|G|$. Thus for any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - G| > \varepsilon) = \mathbb{P}(|G| > \varepsilon/2) > 0 \quad \text{for every odd } n.$$

For even n the probability is zero. It therefore does not converge to zero, and convergence in probability to G fails. □

High priority *Exercise 3: The CLT for a sample proportion*

Exercise 3. For i.i.d. $B_i \sim \text{Bernoulli}(p)$, identify a standard Normal limit for the sample proportion when $0 < p < 1$. Explain the boundary cases $p = 0, 1$ and write a corresponding approximate probability for $|\overline{B}_n - p| \le z\sqrt{p(1-p)/n}$, with fixed $z > 0$.

Solution. Because $B_i^2 = B_i$, its mean is p and variance is $p - p^2 = p(1-p)$. The CLT gives

$$T_n = \frac{\sqrt{n}(\overline{B}_n - p)}{\sqrt{p(1-p)}} \Rightarrow N(0,1).$$

Both endpoints $-z, z$ are continuity points of the Normal CDF, so $\mathbb{P}(-z \le T_n \le z) \to \Phi(z) - \Phi(-z) = 2\Phi(z) - 1$. For completeness, a possible atom at $-z$ vanishes: for every $\delta > 0$, its limsup is at most $\Phi(-z) - \Phi(-z - \delta)$, then let $\delta \downarrow 0$. This justifies the closed lower endpoint. Consequently the stated probability is approximately $2\Phi(z) - 1$ for large n. If $p = 0$ or $p = 1$, every observation equals p almost surely, and the centered sample mean is exactly zero. There is a degenerate limit, and dividing by $\sqrt{p(1-p)}$ is not defined. □

High priority *Exercise 4: Scalar projections and zero variance*

Exercise 4. Let $X \in \mathbb{R}^d$ have mean μ and finite second moments, with covariance Σ. For fixed $c \in \mathbb{R}^d$, prove that $c^\top X$ has finite second moment, find its mean and variance, and treat $c^\top \Sigma c = 0$. Explain the vector CLT reduction.

Solution. Cauchy–Schwarz gives $|c^\top X|^2 \le \|c\|^2 \|X\|^2$, whose expectation is finite. Linearity gives $\mathbb{E}(c^\top X) = c^\top \mu$, and

$$\text{Var}(c^\top X) = \mathbb{E}[c^\top (X - \mu)(X - \mu)^\top c] = c^\top \Sigma c.$$

If this variance is zero, the nonnegative random variable $[c^\top(X-\mu)]^2$ has expectation zero, hence is zero almost surely. For a positive projected variance, the scalar CLT applies to i.i.d. $c^\top(X_i-\mu)$. For a zero projected variance the centered sum is zero almost surely. In both cases the limit law is $N(0, c^\top \Sigma c)$, which is the projection of $N(0,\Sigma)$. Cramér–Wold now proves the vector statement, without an invertibility assumption on Σ. □

Low priority *Exercise 5: The fourth moment of a sum*

Exercise 5. Let X_i be i.i.d., centered, with variance one and finite fourth moment $m_4 = \mathbb{E}\, X_1^4$. For $S_n = n^{-1/2}\sum_i X_i$, calculate $\mathbb{E}\, S_n^4$ exactly and identify its limit. Apply the formula to signs $X_i \in \{-1, 1\}$ with equal probabilities.

Solution. Only multiplicity patterns 4 and $2+2$ survive: every other pattern has a singleton index, which contributes a zero mean by independence. There are n fourth-power terms. For each pair $i<j$, there are $4!/(2!2!) = 6$ orders of the two indices. Therefore

$$\mathbb{E}\, S_n^4 = \frac{1}{n^2}\left(n m_4 + 6\binom{n}{2}\right) = 3 + \frac{m_4 - 3}{n} \longrightarrow 3 = \mathbb{E}\, G^4.$$

For the signs, $m_4 = 1$, so $\mathbb{E}\, S_n^4 = 3 - 2/n$. In particular, at $n = 4$ the value is $5/2$, also obtained directly from the five probabilities in the main example. A finite-sample fourth moment need not equal its limiting Gaussian value. □

Mid priority *Exercise 6: Continuity points in both tails*

Exercise 6. Show that a CDF has at most countably many discontinuities and justify the simultaneous choice of $-M, M$ in part (3) of Theorem 1.

Solution. The jump at x is $F(x) - F(x-) = \mathbb{P}(X = x)$, by continuity of probability from below. For each positive integer r, there are at most r points with mass at least $1/r$, since total mass is one. Every positive mass belongs to one of these finite sets; their countable union is the set of discontinuities D. Given $\varepsilon > 0$, choose R so large that $\mathbb{P}(|X| \geq R) < \varepsilon/2$; the events $\{|X| \geq R\}$ decrease to the empty set as $R \to \infty$. Choose $M > R$ outside the countable set $D \cup (-D)$. Both $-M$ and M are continuity points, and

$$1 - F(M) + F(-M) = \mathbb{P}(X > M) + \mathbb{P}(X \leq -M) \leq \mathbb{P}(|X| \geq R) < \varepsilon/2.$$

Avoiding just the discontinuities at the positive endpoint would not justify convergence at the negative endpoint as well. □

Mid priority *Exercise 7: Continuous approximations to an indicator*

Exercise 7. Assume $\mathbb{E}\, h(X_n) \to \mathbb{E}\, h(X)$ for every bounded continuous h on $\mathbb{R}$. For a continuity point x of F, use linear ramps of width δ to prove $F_n(x) \to F(x)$. Also verify the reverse implication for variables supported on one common finite set.

Solution. Let h_- equal one on $(-\infty, x-\delta]$, decrease linearly to zero on $[x-\delta, x]$, and vanish above x. Let h_+ equal one on $(-\infty, x]$, decrease linearly to zero on $[x, x+\delta]$, and vanish above $x+\delta$. Both are bounded and continuous, and $h_- \leq \mathbf{1}_{(-\infty,x]} \leq h_+$. Taking expectations and limits gives

$$F(x-\delta) \leq \mathbb{E}\, h_-(X) \leq \liminf_n F_n(x) \leq \limsup_n F_n(x) \leq \mathbb{E}\, h_+(X) \leq F(x+\delta).$$

Let $\delta \downarrow 0$ and use continuity at x. For the reverse direction on a common finite support $\{a_1 < \cdots < a_m\}$, choose $b_0 < a_1$, $b_m > a_m$, and $b_j \in (a_j, a_{j+1})$. All b_j are continuity points of the limit CDF, so $\mathbb{P}(X_n =$

$a_j) = F_n(b_j) - F_n(b_{j-1}) \to \mathbb{P}(X = a_j)$. For any bounded continuous h, sum $h(a_j)\,\mathbb{P}(X_n = a_j)$ over the finite set to obtain expectation convergence. □

Mid priority *Exercise 8: Derivatives of a characteristic function*

Exercise 8. For $\mathbb{E}\,X^2 < \infty$, justify $\psi'(t) = \mathbb{E}[iXe^{itX}]$ and $\psi''(t) = -\,\mathbb{E}[X^2 e^{itX}]$. Explain the use of DCT for complex-valued expressions and recover the first two derivatives at zero.

Solution. The inequality $|e^{iv} - 1| \le |v|$ for real v follows by integrating ie^{iu} from 0 to v. Consequently

$$\left|\frac{e^{i(t+h)X} - e^{itX}}{h}\right| \le |X|.$$

This bound is integrable because $\mathbb{E}\,|X| \le (\mathbb{E}\,X^2)^{1/2}$. The difference quotient converges pointwise to iXe^{itX}; apply DCT to real and imaginary parts to obtain $\psi'(t)$. Its difference quotient is bounded by X^2, so a second application gives $\psi''(t) = -\,\mathbb{E}[X^2 e^{itX}]$. DCT also shows the derivatives are continuous. At zero, $\psi(0) = 1$, $\psi'(0) = i\,\mathbb{E}\,X$, and $\psi''(0) = -\,\mathbb{E}\,X^2$. Centering removes the first-order term; the second-order coefficient is negative, not positive. □

Mid priority *Exercise 9: Fixed affine transformations*

Exercise 9. If $X_n \Rightarrow X$ in $\mathbb{R}^d$, prove $AX_n + b \Rightarrow AX + b$ for a fixed matrix $A \in \mathbb{R}^{k\times d}$ and a fixed vector $b \in \mathbb{R}^k$. Does this require A to be invertible?

Solution. For any $h \in C_b(\mathbb{R}^k)$, define $g(x) = h(Ax + b)$. The affine map is continuous, so $g \in C_b(\mathbb{R}^d)$. Hence

$$\mathbb{E}\,h(AX_n + b) = \mathbb{E}\,g(X_n) \longrightarrow \mathbb{E}\,g(X) = \mathbb{E}\,h(AX + b).$$

The test-function criterion proves the claimed weak convergence. Neither injectivity nor invertibility is needed. This includes constant maps, centering, fixed rescaling, and projection onto a lower-dimensional space. A matrix depending on n requires an additional argument. □

Low priority *Exercise 10: The boundary term in a tail moment*

Exercise 10. Derive (6) for $Y \ge 0$, $N > 0$, integer $k \ge 1$, allowing infinite expectations. Explain why deleting its first term is wrong. Apply the identity to a uniform subgaussian tail bound.

Solution. For each realized value $y \ge 0$,

$$y^k \mathbf{1}\{y > N\} = N^k \mathbf{1}\{y > N\} + \int_N^\infty kt^{k-1}\mathbf{1}\{y > t\}\,dt.$$

Both sides vanish for $y \le N$, and for $y > N$ the integral equals $y^k - N^k$. Tonelli gives the expectation identity. For a constant $Y = N + \delta$, the integral alone is $(N+\delta)^k - N^k$, which can be arbitrarily small as $\delta \downarrow 0$ although the left side approaches N^k. Thus no universal multiplicative constant repairs that omission for all such Y. If $\mathbb{P}(Y_n > t) \le Ce^{-ct^2}$ uniformly in n, the missing boundary term is at most $CN^k e^{-cN^2} \to 0$. The integral remainder is at most $C\int_N^\infty kt^{k-1}e^{-ct^2}\,dt \to 0$, as the integrand is integrable. Together these prove uniform decay of the tail moments. □

Low priority *Exercise 11: A tail bound for bounded summands*

Exercise 11. Suppose the independent variables X_i have mean zero and $|X_i| \le M$, where $M > 0$. Show that $S_n = n^{-1/2}\sum_{i=1}^n X_i$ satisfies $\mathbb{P}(|S_n| \ge t) \le 2e^{-t^2/(2M^2)}$ for $t > 0$, uniformly in n.

Solution. Convexity on $[-M, M]$ yields, for real s,

$$e^{sx} \le \frac{M+x}{2M} e^{sM} + \frac{M-x}{2M} e^{-sM}.$$

Take expectations and use $\mathbb{E}\, X_i = 0$ to get $\mathbb{E}\, e^{sX_i} \le \cosh(sM) \le e^{s^2M^2/2}$. The last inequality follows by comparing power series, since $(2j)! \ge 2^j j!$. Independence therefore gives $\mathbb{E}\, e^{sS_n} \le e^{s^2M^2/2}$. For $s > 0$, Markov's inequality gives $\mathbb{P}(S_n \ge t) \le e^{-st+s^2M^2/2}$. Choose $s = t/M^2$ to obtain $e^{-t^2/(2M^2)}$. Apply the same argument to $-S_n$ and add the two bounds. These constants do not depend on n, proving the uniform subgaussian property needed in the moment proof. □

Low priority *Exercise 12: Checking the scale in Normal approximation*

Exercise 12. For a centered variable with variance $\sigma^2 > 0$ and $\rho = \mathbb{E}\,|X|^3 < \infty$, prove $\rho \ge \sigma^3$ and calculate $\mathbb{E}\,|G_\sigma|^3$ for $G_\sigma \sim N(0, \sigma^2)$. Show why Berry–Esseen uses ρ/σ^3, and specialize the smooth and CDF bounds to $X = \pm 2$ with equal probabilities.

Solution. Apply Jensen to $u^{3/2}$ and $U = |X|^2$: $\rho = \mathbb{E}\, U^{3/2} \ge (\mathbb{E}\, U)^{3/2} = \sigma^3$. For $G \sim N(0,1)$, symmetry and the change of variable $u = x^2/2$ give

$$\mathbb{E}\,|G|^3 = \frac{2}{\sqrt{2\pi}} \int_0^\infty x^3 e^{-x^2/2}\, dx = \frac{4}{\sqrt{2\pi}} \int_0^\infty u e^{-u}\, du = 2\sqrt{2/\pi}.$$

Thus $\mathbb{E}\,|G_\sigma|^3 = 2\sqrt{2/\pi}\,\sigma^3$. The standardized input X/σ has variance one and third absolute moment ρ/σ^3. Replacing X by aX, $a \ne 0$, multiplies both ρ and σ^3 by $|a|^3$, leaving their ratio unchanged. For $X = \pm 2$, $\sigma^2 = 4$ and $\rho = 8$. The smooth bound compares S_n with $N(0,4)$ and is at most $8(1 + 2\sqrt{2/\pi})B/(6\sqrt{n})$. The CDF bound compares $S_n/2$ with $N(0,1)$ and is at most $C_{\mathrm{BE}}/\sqrt{n}$, or equivalently compares $\mathbb{P}(S_n \le a)$ with $\Phi(a/2)$ at that same rate. □

Low priority *Exercise 13: Scheffé's theorem and loss of mass*

Exercise 13. Prove Theorem 8 using $\min(f_n, f)$. Explain the necessity of $\int f\, d\nu = 1$ by considering $f_n(x) = \mathbf{1}_{[n,n+1]}(x)$ on $\mathbb{R}$. Verify the factor $1/2$ in total variation distance for two densities.

Solution. We have $\min(f_n, f) \to f$ almost everywhere and $0 \le \min(f_n, f) \le f$, with f integrable. DCT gives $\int \min(f_n, f)\, d\nu \to 1$. Since $|a - b| = a + b - 2\min(a, b)$ for nonnegative a, b,

$$\int |f_n - f|\, d\nu = 2 - 2 \int \min(f_n, f)\, d\nu \longrightarrow 0.$$

For the shifting interval densities, $f_n(x) \to 0$ at every fixed x, but $\int f_n\, dx = 1$. The zero limit is not a probability density, and $\int |f_n - 0|\, dx = 1$; probability mass escapes to infinity. Finally set $h = f - g$. Because $\int h = 0$, $\int h^+ = \int h^- = \frac{1}{2}\int |h|$. For every measurable A, $-\int h^- \le \int_A h \le \int h^+$, with the upper endpoint attained at $A = \{h > 0\}$. This proves $\sup_A |\int_A (f - g)| = \frac{1}{2}\int |f - g|$. □

References

[1] Rick Durrett. *Probability: theory and examples*, volume 49. Cambridge university press, 5th edition, 2019.

[2] Jun Shao. *Mathematical Statistics*. Springer, 2nd edition, 2003.

[3] Terence Tao. *Topics in random matrix theory*, volume 132. American Mathematical Soc., 2012.

What changed

October 19, 2026: Corrected notation, wording and cross-references after the course-wide errata review; refreshed the linked HTML and PDF editions.

October 14, 2026. Added issue-reporting links to the webpage and PDF. The mathematics and priority assignments are unchanged in this update.